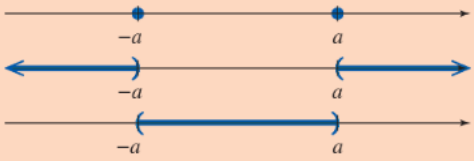
MATH 1075 - Section 9.4 Worksheet

10/27/2026

1 Absolute Value Inequalities

Solving Absolute Value Equations and Inequalities

Let a be a real number such that $a > 0$. Then

Equation/Inequality	Solution (Equivalent Form)	Graph
$\|x\| = a$	$x = -a$ or $x = a$	
$\|x\| > a$	$x < -a$ or $x > a$	
$\|x\| < a$	$-a < x < a$	

Solve the absolute value intequalities.

1. $|x| = 5, |x| < 5, |x| > 5$

2. $|x| = -5, |x| < -5, |x| > -5$

3. $5 \leq |2x - 1|$

4. $12 \leq |9 - 4x| - 2$

5. $4 > -1\left|\frac{2x+1}{4}\right|$

6. $8 < |4 - 3x| + 12$

2 Absolute Value Inequalities, alternative method

Solving Inequalities by Using the Test Point Method

Step 1 Find the boundary points of the inequality. (Boundary points are the real solutions to the related equation and points where the inequality is undefined.)

Step 2 Plot the boundary points on the number line. This divides the number line into intervals.

Step 3 Select a test point from each interval and substitute it into the original inequality.

- If a test point makes the original inequality true, then that interval is part of the solution set.

Step 4 Test the boundary points in the original inequality.

- If the original inequality is strict ($<$ or $>$), do not include the boundary points in the solution set.
- If the original inequality is defined using $\leq$ or $\geq$, then include the boundary points that are defined within the inequality.

Note: Any boundary point that makes an expression within the inequality undefined must *always* be excluded from the solution set.

Solve the absolute value intequalities.

1. $x + 6 < -2$ and $5x < 30$

2. $5x + 2x \geq -21$ and $-9x + 3x \geq -24$

3. $\frac{5}{2}(x + 2) < -6$ and $\frac{3}{4}(x - 2) < 1$

4. $-8 \geq -3x - 2$ and $3(x - 7) + 16 > 4x$